

Embodied Learning for Deaf Signers: Exploring the Potential of XR and Haptic Technologies

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ABSTRACT

Deaf and Hard of Hearing (DHH) individuals face barriers to educational access. While accustomed to visual information, visual channel oversaturation can lead to classroom overwhelm. A focus group study solicited opinions on several technologies that could enhance DHH learning, including an extended reality device, a haptic vest, and wristbands. Participants were generally positive about these, but noted shortcomings and ethical concerns that needed to be addressed before they could be practical for DHH individuals in learning contexts. Overall, our work explores these technologies with the DHH community and aims to enhance them to better support DHH STEM learners.

Index Terms: Deaf, DHH, Multimodality, XR, Haptics, STEM learning, AR.

1 INTRODUCTION AND RELATED WORK

Deaf and Hard of Hearing (DHH) people in America have historically been marginalized from STEM education and professions, with gaps in attainment and larger deaf–hearing educational disparities in STEM than in non-STEM fields [1]. There is a critical need for research-driven, innovative STEM learning approaches that address the specific linguistic, cognitive, and educational demands faced by DHH learners as well as their desires for innovation and pragmatics in these spaces.

Extended reality (XR) and vibrotactile (VT) haptics offer new opportunities for multimodal STEM learning, including simulated, hands-on training that supports workforce development [2, 3, 4, 5]. Yet, despite rapid advances in these technologies, there is no empirical understanding of how to effectively integrate different information channels, such as immersive or pass-through video, English captions, sign language avatars, and haptic feedback, into learning environments. Key questions remain about the conditions under which each modality provides benefits, how these modalities interact, and the usability challenges that may impede adoption. XR and haptic technologies are emerging, offering underexplored opportunities in this space through their unique multimodal channels. To date, no research has directly probed the use of these technologies with DHH STEM content learners.

This paper reports the first step toward characterizing the challenges and opportunities of XR-based STEM learning for DHH adults. We conducted an in-depth focus group with six fluent ASL-signing DHH participants. The primary contributions of this paper are:

1. A characterization of DHH adults' reactions to XR and haptic technologies, capturing both enthusiasm for learning tools that leverage 3D and embodied interaction, and skepticism

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about the current use of haptics for conveying educational content.

2. A set of thematic insights—Physicality, Novelty, Connection, Overwhelm, Learning Potential, and Accessibility—that map barriers to XR and VT haptic adoption by DHH adults, and how they interpreted multimodal learning experiences.
3. A roadmap for research and development, identifying priorities and methodological considerations for future multimodal learning studies.

2 METHODOLOGY

We adopted an exploratory qualitative design, focusing on augmented reality delivered via the Apple Vision Pro (AVP) and haptic feedback systems, namely the bHaptics TactSuit and TactSleeves. We conducted a focus group, beginning with an introduction to haptics and XR, followed by experiences with the technologies. We divided participants into three pairs, and each pair rotated through four stations: the AVP station, the Haptics station, the Interview station, and a Break station. Each pair did not participate in the Interview station until they had passed the AVP and the Haptics stations. All activities took place directly in ASL, with native and fluent-signing research team members.

2.1 Equipment

Extended Reality (XR) device: We used two Apple Vision Pro (AVP) 256GB devices equipped with head-mounted augmented reality capabilities, each loaded with four sample STEM learning apps (e.g., solar system, human biology). Some of these apps were fully immersive, while others used pass-through AR elements.

Haptics: At the vibro-tactile (VT) haptics station, participants used the bHaptics TactSuit Pro (a 32-motor haptic vest) and a Tact-Sleeve (three haptic motors) on each wrist. Participants wore the VT haptic equipment while watching a custom video with corresponding haptic signals prepared by the research team. We designed the video and haptic signals to highlight possible use cases of VT haptic devices, as detailed in Figure 1. While not directly showing STEM materials, the videos and haptics highlighted how haptics could communicate relevant information such as repeated rhythms or getting a user's attention.

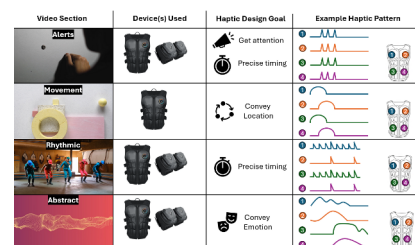


Figure 1: The four segments used in the VT haptics demonstration video. The example haptic patterns show the signal sent to individual haptic motors over time.

Questionnaires: A pre-questionnaire was used to gather attitudes toward technology, captions, signing avatars, haptics, and emerging technologies. A post-questionnaire included the same questions. All questions for opinion data were asked in the form of a statement, to which participants responded along a 5-point Likert scale (1="Strongly Disagree," 5="Strongly Agree," 3="Neutral").

3 RESULTS

Six deaf adults (ages 24-34, $M=27.38$, $SD=3.86$) participated in a paid, IRB-approved, four-hour session and provided informed consent prior to participation.

Quantitative Results: Paired t-tests were conducted on the pre- and post-questionnaires to see if ratings changed over time. As shown in Table 1, there was no statistical difference; however, overall numerical scores increased, suggesting a potential rise in interest in the technologies.

Table 1: Pre-Post Changes in Technology Attitudes

Measure (out of 5=strongly agree)	Pre (M, SD)	Post (M, SD)	t(5), p
XR composite (out of 15)	12.17 (1.84)	13.33 (1.97)	-1.58, 0.18
3D educational materials	4.33 (0.82)	4.67 (0.52)	-1, 0.36
Avatar interest	3.83 (1.17)	3.83 (1.47)	0, 1.00
Comm. support gloves	2.00 (0.89)	2.17 (1.17)	-1, .36
Comm. support sign gen.	3.67 (1.21)	4.17 (1.33)	-1.46, .20
Haptics language	3.33 (1.03)	3.83 (0.98)	-2.24, 0.07
Haptics spatial	3.17 (1.47)	4.33 (0.52)	-1.78, 0.13
New tech comfort	4.17 (0.75)	4.67 (0.52)	-1.46, 0.20

Qualitative Results: Initial codes were generated from participants' comments for each interview prompt. Related codes were then clustered into higher-level categories to capture shared patterns. Through a review of these categories, themes were identified that characterize recurring ideas across the dataset. This yielded six overarching themes, described here.

Physicality: Consensus that the technologies were not as easy or comfortable to use (i.e., heavy and needing to recalibrate eye-tracking for each person) as would be necessary for regular interaction. Suggestions for smaller future iterations, with a preference for the haptic vest over the wrist-worn devices, given the limited gain.

Novelty: New technologies brought both issues and joys; while several were excited by XR technology, there were ethical concerns (e.g., deployment of immature technology). Despite the difficult setup, participants expressed positive first impressions. **Connection:** Participants specifically noted that, while XR and haptics could bridge gaps in accessibility, it might be difficult to distinguish between interpersonal and device-driven interactions. Furthermore, participants envisioned the shared use of XR technologies in the classroom. **Overwhelm:** Potential for competing streams of information (i.e., media, signing, captions, haptics), as being demanding, over-saturating not only the visual modality but also the tactile modality. Customization was mentioned as a tool to address increased cognitive demands. **Learning Potential:** XR might be well-suited for DHH learners and STEM content, particularly for objects hard to manipulate and topics containing spatial information that can be clearly visualized (e.g., cell layers, dissections, 3D math). Haptic technologies were preferred when the connection to visual stimuli was apparent. **Accessibility:** Participants identified ways that the devices could enhance their access to information or remove barriers, e.g., the VT haptic vest's potential to support vestibular balance or attentional cuing, or the AVP's potential to use signed language input.

4 DISCUSSION AND CONCLUSION

Overall, participants showed cautiously optimistic reactions to XR, recognizing its potential to make STEM learning more engaging

and accessible. Participants noted haptics can convey certain types of information effectively, but expressed skepticism that they can make a meaningful contribution to learning. While there was openness in seeing how the technology develops, a common thread was that several aspects of the technology need to be worked out before they can be readily implemented into daily life.

Participants were slightly, but not significantly, more positive about XR for STEM learning following the hands-on explorations (Table 1). Participants expressed frustration with the XR device setup and use, which pose significant barriers to conducting this type of research and to adopting AVPs for classroom use. Our results suggest that haptic technologies may be beneficial for XR learning, dependent on the content and context.

One motivation for this work was examining how multimodal learning technologies may affect cognitive overload. Without prompting, focus group participants mentioned how XR and haptic technologies could become overwhelming or distracting. During the group discussion, participants suggested that being able to customize the available information would alleviate cognitive demand.

5 LIMITATIONS AND FUTURE DIRECTIONS

Our study included only six participants, so future research will need to draw more widely applicable conclusions from larger sample sizes. Our participants had relatively brief, passive experiences with XR and haptics, with limited opportunities for customization. It will be crucial to develop prototypes, in partnership with DHH learners, that integrate sign language content, haptics, and captions into XR learning applications. For instance, the AVP's capabilities could be utilized to accept signed language input. With such prototypes, it will be feasible to more rigorously assess how DHH learners balance visual and tactile channels to perceive and learn with XR tools.

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